

NL Question Q

OFFLINE PRE-PROCESSING (zero API cost)

Phase 1: CHESS Pruning FAISS Top- k retrieval

$S' \subseteq S$

Phase 2: MCI-SQL Enrichment MIN/MAX, cardinality, samples

C_{meta}

Phase 3: Context Assembly DDL + MCI JSON + MAGIC guidelines

P_{enriched}

ASYMMETRIC LLM LOOP (≤ 3 API calls)

Phase 4a: Generator openai/gpt-oss-120b via Groq

y_0

Phase 4ab: Reflector meta-llama/llama-4-scout-17b

Logic mismatch

OK

Semantic Validation

Syntax/Empty/Null err

Phase 4c: Resilient Critic openai/gpt-oss-20b

-PASS

y_{i+1}

Final SQL $\hat{y}$